

PinProbe A1

Box Control SCPI Command Reference

Communication Commands / Device Control / Error Handling

Customer Integration Material

SCPI / UART

2026-09-24

Document updated: 2026-09-24. The IO probe selector and persistent input/output trigger configuration are included for host integration and maintenance reference. Existing applications remain compatible with the default `LEVEL` behavior.



Device Information

Item	Default Value	Description
Manufacturer	GTS	*IDN? field 1; can be changed with SYSTem:IDN1
Model	PINPROBEA1	*IDN? field 2; can be changed with SYSTem:IDN2
Serial Number	20250626	*IDN? field 3; can be changed with SYSTem:IDN3
IDN Firmware Version	V0.0.9	*IDN? field 4; can be changed with SYSTem:IDN4
Compiled Firmware Version	v1.0.0+<build-id>	SYSTem:VERSion? returned

*IDN? returns the current SCPI IDN configuration: Manufacturer,Model,Serial Number,IDN Firmware Version The IDN fields are stored in device configuration and may differ from compiled firmware version.

Communication Conventions

Item	Description
Serial Baud Rate	115200
Command Terminator	\r\n
Normal Response	Single-line text
Batch Log Response	READ:LOG:ALL? may return multi-line text
OTA Data Block	Uses SCPI definite-length binary block; max payload per block is 128 bytes

SCPI errors can come from parser error queue, command responses beginning with ERR... text, and device exception logs. Host software should process both command responses and logs; see Error and Exception Handling.

Basic Commands (IEEE 488.2)

Command	Parameter	Response	Description
*CLS		None	Clear status and error queue
*IDN?		GTS,PINPROBEA1,20250626,V0.0.9	Query device identification
*RST		None	Reset SCPI parser state
*STB?		Status byte	Query status byte
*WAI		None	Wait for command completion
*OPC?		1	Query operation-complete status

System Commands

Command	Parameter	Response	Description
SYSTem:ERRor[:NEXT]?		Error code and description	Query next error message
SYSTem:ERRor:COUNT?		Error count	Query error queue count
SYSTem:VERSion?		v1.0.0+<build-id>	Query compiled firmware version and build ID
SYSTem:UPTime?		Seconds	Query system uptime
SYSTem:REBoot		OK	Software reset after about 200 ms
SYSTem:FLASH:ID?		OK ID=<JEDEC> SR1=<xx> SR2=<xx> or ERR ...	Query W25Q128 Flash ID and status registers

Example

```
SYSTem:ERRor:NEXT?  
SYSTem:ERRor:COUNT?  
SYSTem:VERSion?  
SYSTem:UPTime?  
SYSTem:FLASH:ID?  
SYSTem:REBoot
```

Device Identification Configuration Commands

Command	Parameter	Response	Description
SYSTem:IDN1	<string>	None	Configure manufacturer field and save to Flash
SYSTem:IDN1?		String	Query manufacturer field
SYSTem:IDN2	<string>	None	Configure model field and save to Flash
SYSTem:IDN2?		String	Query model field
SYSTem:IDN3	<string>	None	Configure serial number/date field and save to Flash
SYSTem:IDN3?		String	Query serial number/date field
SYSTem:IDN4	<string>	None	Configure IDN firmware version field and save to Flash
SYSTem:IDN4?		String	Query IDN Firmware VersionField

Example

```
SYSTem:IDN1 "GTS"  
SYSTem:IDN2 "PINPROBEA1"  
SYSTem:IDN3 "20260727"  
SYSTem:IDN4 "V0.0.9"  
*IDN?
```

Communication Configuration Commands

Command	Parameter	Response	Description
CONFigure:BAUDrate	115200	115200 baudrate is enable	Configure BSM communication baud rate. Only 115200

Example

```
CONFigure:BAUDrate 115200
```

Door and USB Control Commands

Command	Parameter	Response	Description
CONFigure:CYLInder1	OPEN / CLOSE	OPEN / CLOSE	Open or close the box door
READ:CYLInder1:STATE?		Actuator State	Query the current door actuator state
CONFigure:CYLInder2	OPEN / CLOSE	OPEN / CLOSE	Manual USB: CLOSE =insert/connect; OPEN =unplug/retract
READ:CYLInder2:STATE?		Actuator State	Query the current USB actuator state
CONFigure:USB:AUTO	OFF / ON	OFF / ON	Factory USB auto setting; do not modify
READ:USB:AUTO?		OFF / ON	Query the factory USB auto setting
CONFigure:DUT:AUTO	OFF / ON	OFF / ON	Set whether DUT presence detection participates in the USB automatic sequence; saved to Flash
READ:DUT:AUTO?		OFF / ON	Query whether DUT presence detection participates in the USB automatic sequence

USB state naming: CYLInder2 CLOSE =insert/connect; OPEN =unplug/retract.

Actuator State

Return Value	Description
CLOSE	Command accepted; USB inserted/connected
OPEN	Command accepted; USB unplugged/retracted
CLOSING	Closing; USB insertion in progress
OPENING	Opening; USB retraction in progress
CLOSED	Closed; USB inserted/connected
OPENED	Opened; USB unplugged/retracted
CYL_ERR	Actuator error

USB Automatic Sequence

CONFigure:USB:AUTO ON links USB insertion/retraction to door operation; CONFigure:DUT:AUTO ON adds DUT presence gating to that automatic flow.

Scenario	Automatic Action
USB:AUTO OFF	Bypass USB automation and DUT presence detection; run the normal door sequence
USB:AUTO ON and DUT:AUTO OFF	DUT detection does not participate; button confirmation starts USB automatic insertion directly
USB:AUTO ON and DUT:AUTO ON	Check dut_sensor first; insert USB only after DUT is in position, then close after USB insertion is confirmed
DUT not in position	Do not insert USB or close; keep waiting. READ:DUT:STATE? still reports the actual sensor state
USB insertion fault	Do not close; log USB_INSERT_FAIL; retract USB if the door is still open; fast-flash red
COMPLETE; open-door button pressed	Open door; after IDLE, retract USB if still inserted
USB retraction fault	Log USB_RETRACT_FAIL; fast-flash red; return IDLE if needed

CONFigure:USB:AUTO is factory-set; **do not modify**. CONFigure:DUT:AUTO is saved to Flash and is effective only when USB:AUTO ON. Query commands are not bypassed: READ:DUT:STATE? always reads the actual dut_sensor.

Lock and Indicator Commands

Command	Parameter	Response	Description
CONFigure:LOCK	UNLOCK / LOCKED	UNLOCK / LOCKED	Set the device lock state
READ:LOCK:STATE?		Lock State	Read the current lock state

Lock State

Parameter / Return Value	Description
UNLOCK	Unlocked
LOCKED	Locked
LOCK ERR	Lock error

Example

```
CONFigure:LOCK LOCKED
CONFigure:LOCK UNLOCK
READ:LOCK:STATE?
```

LED Indicator Commands

Command	Parameter	Response	Description
CONFigure:LED	OFF / GREEN / RED / YELLOW	LED State	Set the LED output state
READ:LED:STATE?		LED State	Read the current LED output
CONFigure:LED:MAP	<i05>,<i06>,<i07>	G,R,Y format	Map IO5/IO6/IO7 to LED colors and save
READ:LED:MAP?		G,R,Y format	Read the IO5/IO6/IO7 LED color map

LED State

Parameter / Return Value	Description
OFF	LED off
GREEN	Green LED
RED	Red LED
YELLOW	Yellow LED
LED ERR	LED error

LED Mapping Parameters

Parameter	Description
G / GREEN	Map this IO line to the green LED
R / RED	Map this IO line to the red LED
Y / YELLOW	Map this IO line to the yellow LED

CONFigure:LED:MAP needs three unique values and must include green, red, and yellow.

Example

```
CONFigure:LED GREEN
CONFigure:LED OFF
READ:LED:STATE?
CONFigure:LED:MAP G,R,Y
READ:LED:MAP?
```

IO Probe Modes

Command	Parameter	Response	Description
CONFigure:IO:PROBe	RAW / FORWARD	RAW / FORWARD	Select the IO source used by READ:IO:ALL?; runtime only
READ:IO:PROBe?		RAW / FORWARD	Query the current IO probe mode
READ:IO:ALL?		IN:0xHH,0xHH OUT:0xHH,0xHH	Query all input and output states using the selected probe mode

RAW reports the physical IO sample. FORWARD reports the state-management layer result, including input latches and output forwarding. The boot default is RAW; the probe mode is not saved to Flash.

Input / Output Point Trigger Types

Command	Parameter	Description
CONFigure:INPut#:TYPE READ:INPut#:TYPE?	LEVEL / PULSE # = 1..16	Configure or query an input trigger type; saved to Flash. All inputs default to LEVEL.
CONFigure:OUTPut#:TYPE READ:OUTPut#:TYPE?	LEVEL / PULSE # = 1..16	Configure or query an output trigger type; saved to Flash. All outputs default to LEVEL.
CONFigure:OUTPut:PULSe:WIDTh READ:OUTPut:PULSe:WIDTh?	1..60000 ms	Configure or query the common output pulse width; saved to Flash.

LEVEL forwards the current level directly. PULSE uses the state-management latch/pulse layer and is controlled by the point configuration independently of Risk Mode. For built-in door/USB position paths, the first allowed-direction rising edge sets the latch and a new rising edge during reverse movement clears it; direction change alone does not clear it. Points without an application path retain raw semantics. Risk Mode remains limited to the pressure-sensor close confirmation path, while existing applications remain compatible with the default LEVEL configuration.

The desktop Tools window exposes two IO tabs: Probe selects RAW or FORWARD and displays the values; Configure uses colored indicator lamps to show and change each input/output point type and set the output pulse width.

Example

```
CONFigure:IO:PROBe FORWARD
READ:IO:PROBe?
READ:IO:ALL?
CONFigure:INPut1:TYPE PULSE
CONFigure:OUTPut2:TYPE LEVEL
CONFigure:OUTPut:PULSe:WIDTh 100
READ:INPut1:TYPE?
READ:OUTPut:PULSe:WIDTh?
```

System State Query Commands

Command	Parameter	Response	Description
READ:SYSTem:STATe?		System State	Query system state
CONFIgure:IO:PROBe	RAW / FORWARD	RAW / FORWARD	Select the IO source used by READ:IO:ALL? (runtime only)
READ:IO:PROBe?		RAW / FORWARD	Query the current IO probe mode
READ:IO:ALL?		IN:0xHH,0xHH OUT:0xHH,0xHH	Query all input and output states using the selected probe mode
READ:DUT:STATe?		INPOS / OUTPOS	Query the actual DUT presence sensor state

System StateReturn Value

Return Value	Description
LOCK	System locked
IDLE	System idle
READY	System ready
RUNNING	System running
EMERGENCY	Emergency stop triggered
COMPLETE	System operation complete
SYS ERR	System error

Example

```
READ:SYSTem:STATe?  
CONFIgure:IO:PROBe FORWARD  
READ:IO:PROBe?  
READ:IO:ALL?  
READ:DUT:STATe?
```

READ:DUT:STATe? reads dut_sensor: INPOS means DUT in position, and OUTPOS means DUT not in position. This query always remains valid, regardless of USB:AUTO or DUT:AUTO bypass settings.

E-Stop, Risk Mode, and Boot Diagnostics

Command	Parameter	Response	Description
CONFIgure:ESTOP:TYPE	NC / NO	NC / NO	Configure E-stop input type and save to Flash
READ:ESTOP:TYPE?		NC / NO	Query E-stop input type
CONFIgure:RISK:MODE	OFF / ON	OFF / ON	Configure Risk Mode and save to Flash
READ:RISK:MODE?		OFF / ON	Query Risk Mode
CONFIgure:BOOT:DIAG	OFF / ON	OFF / ON	Configure bootloader diagnostic serial output and save to Flash
READ:BOOT:DIAG?		OFF / ON	Query bootloader diagnostic serial output switch

ParameterDescription

Parameter	Description
NC / NO	NC: E-stop normally closed; NO: E-stop normally open
OFF / ON	OFF: disable function; ON: enable function

Example

Control Mode Commands

Switch between local manual control, remote software control, and compatibility mixed control, and configure whether mode switching is enabled.

Command	Parameter	Response	Description
CONFigure:MODE	LOCAL / REMOTE / MIXED	LOCAL / REMOTE / MIXED	Switch the current control mode; runtime only, not saved to Flash
READ:MODE?		LOCAL / REMOTE / MIXED	Query the current control mode
CONFigure:MODE:ENABle	ON / OFF	ON / OFF	Configure whether control-mode switching is enabled; save to Flash
READ:MODE:ENABle?		ON / OFF	Query whether control-mode switching is enabled

Control Mode Permissions

Mode	Physical-button ordinary actions	SCPI ordinary actions	Queries and safety logic
LOCAL	Allowed	Rejected	Always available
REMOTE	Blocked	Allowed	Always available
MIXED	Allowed	Allowed	Always available

LOCAL is local manual control: ordinary door and USB actions are accepted from the physical panel and rejected from SCPI. REMOTE is remote software control: ordinary door and USB actions are accepted from SCPI and physical-button ordinary actions are blocked. MIXED preserves the current behavior by allowing both sources.

Status, alarm, error-queue, log, IO, sensor, version, and diagnostic queries remain available in all three modes. Emergency stop, laser anti-pinch, RS485 fault protection, power-loss retraction/locking, and other safety state-machine behavior are not disabled by the control mode. Lock and unlock use an independent safety/maintenance permission.

Mode-Switch Enable

Value	Description
ON	Allow CONFigure:MODE to switch between LOCAL, REMOTE, and MIXED
OFF	Lock the current mode and reject ordinary mode changes; CONFigure:MODE LOCAL and CONFigure:MODE:ENABle ON remain available as recovery paths

CONFigure:MODE:ENABle is the persistent configuration item and is written to Flash. CONFigure:MODE changes only the current runtime mode and is not restored from Flash. After power-up, the current mode is always MIXED; only mode_enable is restored from Flash. The firmware rejects the unsafe REMOTE + OFF combination so remote control cannot be permanently locked out. The control-mode functions have been implemented and verified in real-device integration tests.

Example

```
CONFigure:MODE:ENABle ON
CONFigure:MODE REMOTE
READ:MODE?
READ:MODE:ENABle?
CONFigure:MODE MIXED
CONFigure:MODE LOCAL
```

OTA Firmware Upgrade Commands

Command	Parameter	Response	Description
SYSTem:OTA:STATus?		STATE:... SIZE:... RECV:...	Query OTA receiving and verification status
SYSTem:OTA:BOOT?		FLASH:... FLAGS:... MAN:...	Query Bootloader and OTA Manifest status
SYSTem:OTA:BEGIN	<size>,<crc32>,<version>,<image_id>	OK or ERR,<code>,<name>	Start one OTA upload
SYSTem:OTA:DATA	<offset>,<n><len><data>	OK,NEXT:<offset> or ERR,<code>,<name>	Write one OTA data block
SYSTem:OTA:END		OK or ERR,<code>,<name>	End OTA data receiving
SYSTem:OTA:VERify?		Status text or error text	Verify OTA image
SYSTem:OTA:COMMit		OK or error text	Commit upgrade request; device resets after success
SYSTem:OTA:ABORt		OK or error text	Abort current OTA process

OTA StatusResponse

SYSTem:OTA:STATus? response format:

```
STATE:<state> SIZE:<bytes> RECV:<bytes> BLOCKS:<recv>/<total> NEXT:<offset> CRC:0x<crc32> VER:0x<version> IMG:<id> ERR:<code>
```

SYSTem:OTA:BOOT? response format:

```
FLASH:<0|1> FLAGS:<0|1> MAN:<0|1> SEQ:<n> ACT:<n> PEND:<slot> PREV:<slot> BSTATE:<n> ATT:<n>/<max> BERR:<code> MSTATE:<state> SIZE:<bytes> CRC:0x<crc32> VER:0x<version> IMG:<id>
```

OTA Example

```
SYSTem:OTA:STATus?
SYSTem:OTA:BEGIN 4096,305419896,65536,1
SYSTem:OTA:DATA 0,#3128<128 bytes binary payload>
SYSTem:OTA:END
SYSTem:OTA:VERify?
SYSTem:OTA:COMMit
```

Error and Exception Handling

SCPI Parser Errors

When parameters are missing/invalid, a command is unknown, Flash save fails, or low-level hardware communication fails, firmware writes to the SCPI error queue. Queue capacity is 17 entries. Read with:

```
SYSTem:ERRor:COUNT?  
SYSTem:ERRor:NEXT?  
*CLS
```

Scenario	Typical Behavior	Recommended Handling
Unknown command	Serial port may output <code>**ERROR: -113, "Undefined header"</code> ; queue may return <code>-113, "Undefined header"</code>	Stop sending this command; check spelling and hierarchy
Missing parameter	<code>-109, "Missing parameter"</code>	Provide required parameter
Ordinary SCPI action rejected in LOCAL	<code>-201, "Invalid while in local"</code>	Switch to REMOTE or MIXED before sending ordinary door or USB actions through SCPI
Illegal parameter value	<code>-224, "Illegal parameter value"</code> or <code>-222, "Data out of range"</code>	Check enum values, duplicate LED mapping, and numeric range
Flash save failure	<code>-320, "Storage fault"</code>	Configuration may not persist; retry and read back the related query command
UART/RS485 hardware or communication failure	<code>-240, "Hardware error"</code> or <code>-360, "Communication error"</code>	Query <code>READ:LOG:STATus?</code> / <code>READ:IO:ALL?</code> , and check IO expansion board link

SCPI error queue is read by dequeue; `SYSTem:ERRor:NEXT?` pops one entry each time. `*CLS` clears status and error queue.

Command Responses Beginning with ERR...

Some commands return errors in normal responses instead of the SCPI error queue: ERR... :

Command	Error Format	Description
<code>SYSTem:FLASH:ID?</code>	<code>ERR INIT=<name></code> or <code>ERR ID=<name></code>	W25Q128 initialization or JEDEC ID read failed
OTA Command	<code>ERR, <code>, <name></code>	OTA process error; see table below

OTA Error Codes

Code	Name	Meaning
0	OK	Success
1	BUSY	OTA process busy
2	BAD_STATE	Current state does not allow this OTA step
3	BAD_PARAM	Parameter error
4	FLASH_FAIL	Flash operation failed
5	SIZE_OVERFLOW	Image size exceeds allowed range
6	CRC_FAIL	CRC verification failed
7	CAN_TIMEOUT	CAN distribution timeout
8	CAN_NACK	CAN node rejected
9	NODE_NOT_READY	Node not ready
10	BOOT_FAIL	Bootloader upgrade failed
11	APP_NOT_CONFIRMED	App not confirmed
12	BLOCK_MISSING	OTA data block missing

1. When command response begins with `ERR` or `ERROR` immediately read `SYSTem:ERRor:NEXT?` until no error remains.
2. Periodically read `READ:LOG:STATus?`; if `PEND` is nonzero, read `READ:LOG:NEXT?` or `READ:LOG:ALL?`.
3. When `[ERROR]` is found, stop automatic testing and collect `READ:SYSTem:STATe?` / `READ:IO:ALL?` / `READ:CYLInder1:STATe?` / `READ:CYLInder2:STATe?`.
4. When `[WARN]` is found, decide by event name; manually confirm USB failures, RS485 faults, and IO write failures before continuing.

Recommended input air pressure: about `0.33 MPa`; set the electronic pressure gauge threshold to about `0.3 MPa`.

Exception	Log Level	Firmware Action	Recovery Recommendation
E-stop <code>ESTOP</code>	<code>[ERROR]</code>	Enter <code>EMERGENCY</code> , red light, open door, and lock system	After resetting E-stop and clearing laser path, press Start again to unlock and recover
Laser anti-pinch <code>LASER</code>	<code>[ERROR]</code>	If obstruction is detected late in closing, enter <code>EMERGENCY</code> , red light, open door, and lock system	Clear obstruction, then press Start again to unlock and recover
RS485 fault <code>RS485_FAULT</code>	<code>[WARN]</code>	Mark IO link unavailable; pause this operation and request yellow alarm	Wait for <code>RS485_RECOVERED</code> , then read IO and system state to confirm
Risk Mode pressure event <code>RISK_PRESSURE</code>	<code>[WARN]</code>	In Risk Mode, use pressure and learned time to determine door-close completion	Check door lower-limit or pressure sensor state
Low air pressure <code>AIR_LOW</code>	<code>[WARN]</code>	Record diagnostic log only; E-stop is not triggered	Wait for <code>clear_after</code> record or check air path
IO write failure <code>IO_WRITE_FAIL</code>	<code>[WARN]</code>	Related actuator command may not actually output	Check IO board communication, power, and actuator wiring
USB insertion failure <code>USB_INSERT_FAIL</code>	<code>[WARN]</code>	USB may retract, fast red flash, or yellow flash in <code>COMPLETE</code>	Query <code>READ:CYLInder2:STATe?</code> , check USB insertion cylinder and position sensor
USB retraction failure <code>USB_RETRACT_FAIL</code>	<code>[WARN]</code>	Fast red flashing; return to <code>IDLE</code>	Query <code>READ:CYLInder2:STATe?</code> , check USB retraction cylinder and position sensor

Debug Switch Commands

Command	Parameter	Response	Description
<code>CONFigure:DEBUg:STATe</code>	<code>OFF</code> / <code>ON</code>	<code>OFF</code> / <code>ON</code>	Enable/disable runtime state-transition debug output
<code>READ:DEBUg:STATe?</code>		<code>OFF</code> / <code>ON</code>	Query state-transition debug-output switch
<code>CONFFigure:DEBUg:ACTIon</code>	<code>OFF</code> / <code>ON</code>	<code>OFF</code> / <code>ON</code>	Enable/disable runtime action-duration debug output
<code>READ:DEBUg:ACTIon?</code>		<code>OFF</code> / <code>ON</code>	Query action-duration debug-output switch
<code>CONFFigure:DEBUg:EVENT</code>	<code>OFF</code> / <code>ON</code>	<code>OFF</code> / <code>ON</code>	Enable/disable runtime event debug output
<code>READ:DEBUg:EVENT?</code>		<code>OFF</code> / <code>ON</code>	Query event debug-output switch
<code>CONFFigure:DEBUg:IO</code>	<code>OFF</code> / <code>ON</code>	<code>OFF</code> / <code>ON</code>	Enable/disable runtime IO debug output
<code>READ:DEBUg:IO?</code>		<code>OFF</code> / <code>ON</code>	Query IO debug-output switch

Debug switches are runtime state; default is `OFF`, not written to Flash.

Example

```
CONFFigure:DEBUg:STATe ON
READ:DEBUg:STATe?
CONFFigure:DEBUg:IO OFF
READ:DEBUg:IO?
```

Device Log Commands

Command	Parameter	Response	Description
CONFigure:LOG:UART	OFF / ON	OFF / ON	Enable/disable live UART logging
READ:LOG:UART?		OFF / ON	Read live UART logging setting
READ:LOG:NEXT?		One log entry or EMPTY	Read the next pending device log
READ:LOG:ALL?		Multi-line logs or EMPTY	Read all pending device logs
READ:LOG:STATUS?		PEND:... CAP:... NEXT:...	Read the log buffer status
CONFigure:LOG:CLEar		OK	Clear the log buffer and counters

Log Status Response

PEND:<pending> CAP:<capacity> NEXT:<next_seq> DROP:<drop_count> RDROP:<realtime_drop_count> UART:<0|1>

Log Line Format

Logs usually start with timestamp and node number:

```
[T+12.345s][N0][INFO][ACTION] CLOSE_DONE duration=300ms
```

Format:

```
[T+<time since boot>][N<node number>][<level>][<module>] <event and parameters>
```

Field	Description
T+	Boot-relative time; restarts after reset
N	Node number; single-device default is N0
INFO	Normal state, action, command, or recovery
WARN	Exception or diagnostic record to check
ERROR	Severe event, e.g. E-stop or laser anti-pinch
STATE / ACTION / EVENT / IO	State, action, event, or IO snapshot

Exception Log Description

Use the log level to judge exceptions. E-stop and laser anti-pinch use [ERROR][EVENT]; other diagnostics usually use [WARN][EVENT]:

Exception Event	Log Example	Meaning and Operation Notes
E-stop triggered	[T+15.120s][N0][ERROR][EVENT] ESTOP close_elapsed=820ms	Opens door and locks; reset E-stop, press Start
Object obstruction	[T+16.020s][N0][ERROR][EVENT] LASER close_elapsed=1720ms	Opens door and locks; clear obstruction, press Start
RS485 fault	[T+12.345s][N0][WARN][EVENT] RS485_FAULT arg0=0 arg1=0	IO link unavailable; wait for recovery log
Risk pressure	[T+16.100s][N0][WARN][EVENT] RISK_PRESSURE close_elapsed=1800ms	Close-pressure condition; check workpiece/resistance
Air low / recovered	[T+20.000s][N0][WARN][EVENT] AIR_LOW low_after=500ms	Low pressure or recovery; check low_after/clear_after
IO write fail	[T+21.000s][N0][WARN][EVENT] IO_WRITE_FAIL target=2 cmd=0x0003	Output write failed; check actuator and IO link
USB insert fail	[T+22.000s][N0][WARN][EVENT] USB_INSERT_FAIL elapsed=2001ms reason=timeout	USB not inserted; check reason cause
USB retract fail	[T+23.000s][N0][WARN][EVENT] USB_RETRACT_FAIL elapsed=2001ms reason=sensor_conflict	USB not retracted; check reason cause

USB Reasons

reason	Description
timeout	USB travel exceeded 2000 ms without reaching position
sensor_conflict	USB upper/lower sensors active at the same time
dual_output	USB insert/retract outputs active at the same time
unknown	Unrecognized reason code

A normal information log is generated when RS485 communication recovers:

```
[T+13.500s][N0][INFO][EVENT] RS485_RECOVERED arg0=0 arg1=0
```

This record means the IO link recovered; the device exits RS485 protection and requests alarm light off. `ESTOP` and `LASER` open the door and lock the system; clear the cause, then press Start. `RISK_PRESSURE` has no separate recovery event; judge from later system, IO, and action logs.

Reading Tips

1. Send `READ:LOG:STATUS?` first to check pending and drop counts.
2. Use `READ:LOG:NEXT?` one by one, or `READ:LOG:ALL?` for all entries.
3. Filter `[WARN]` and `[ERROR]`; after `RS485_FAULT`, wait for `RS485_RECOVERED`.
4. If `DROP` or `RDROP` is nonzero, logs are incomplete; collect system and IO states.

`READ:LOG:NEXT?` and `READ:LOG:ALL?` are dequeue reads; returned records are not repeated. The buffer holds 76 RAM entries and clears after reset or power loss. `DROP` counts overwritten pending records; `RDROP` counts records lost when real-time UART cannot send fast enough. `CONFIGure:LOG:CLEAR` clears logs and both drop counters. With `CONFIGure:LOG:UART ON`, logs appear as unsolicited text on the SCPI port. Keep `OFF` if async text cannot be handled.

Example

```
CONFIGure:LOG:UART ON
READ:LOG:UART?
READ:LOG:STATUS?
READ:LOG:NEXT?
READ:LOG:ALL?
CONFIGure:LOG:CLEAR
```

Command Summary

Category	Command
Basic	<code>*CLS</code> , <code>*IDN?</code> , <code>*RST</code> , <code>*STB?</code> , <code>*WAI</code> , <code>*OPC?</code>
System	<code>SYSTEM:ERROR[:NEXT]?</code> , <code>SYSTEM:ERROR:COUNT?</code> , <code>SYSTEM:VERSION?</code> , <code>SYSTEM:UPTime?</code> , <code>SYSTEM:REBoot</code> , <code>SYSTEM:FLASH:ID?</code>
IDN	<code>SYSTEM:IDN1</code> , <code>SYSTEM:IDN1?</code> , <code>SYSTEM:IDN2</code> , <code>SYSTEM:IDN2?</code> , <code>SYSTEM:IDN3</code> , <code>SYSTEM:IDN3?</code> , <code>SYSTEM:IDN4</code> , <code>SYSTEM:IDN4?</code>
Act.	<code>CONFIGure:CYLInder#</code> , <code>READ:CYLInder#:STATE?</code> , <code>CONFIGure:USB:AUTO</code> , <code>READ:USB:AUTO?</code> , <code>CONFIGure:DUT:AUTO</code> , <code>READ:DUT:AUTO?</code> , <code>CONFIGure:LOCK</code> , <code>READ:LOCK:STATE?</code> , <code>CONFIGure:LED</code> , <code>READ:LED:STATE?</code> , <code>CONFIGure:LED:MAP</code> , <code>READ:LED:MAP?</code>
Status	<code>READ:SYSTEM:STATE?</code> , <code>CONFIGure:IO:PROBe</code> , <code>READ:IO:PROBe?</code> , <code>READ:IO:ALL?</code> , <code>READ:DUT:STATE?</code>
Mode	<code>CONFIGure:MODE</code> , <code>READ:MODE?</code> , <code>CONFIGure:MODE:ENABle</code> , <code>READ:MODE:ENABle?</code>
Cfg.	<code>CONFIGure:BAUDrate</code> , <code>CONFIGure:ESTOP:TYPE</code> , <code>READ:ESTOP:TYPE?</code> , <code>CONFIGure:RISK:MODE</code> , <code>READ:RISK:MODE?</code> , <code>CONFIGure:BOOT:DIAG</code> , <code>READ:BOOT:DIAG?</code> , <code>CONFIGure:INPut#:TYPE</code> , <code>READ:INPut#:TYPE?</code> , <code>CONFIGure:OUTPut#:TYPE</code> , <code>READ:OUTPut#:TYPE?</code> , <code>CONFIGure:OUTPut:PULSe:WIDTh</code> , <code>READ:OUTPut:PULSe:WIDTh?</code>
OTA	<code>SYSTEM:OTA:STATUS?</code> , <code>SYSTEM:OTA:BOOT?</code> , <code>SYSTEM:OTA:BEGIN</code> , <code>SYSTEM:OTA:DATA</code> , <code>SYSTEM:OTA:END</code> , <code>SYSTEM:OTA:VERify?</code> , <code>SYSTEM:OTA:COMMit</code> , <code>SYSTEM:OTA:ABORt</code>
Dbg.	<code>CONFIGure:DEBUg:STATE</code> , <code>READ:DEBUg:STATE?</code> , <code>CONFIGure:DEBUg:ACtion</code> , <code>READ:DEBUg:ACtion?</code> , <code>CONFIGure:DEBUg:EVENT</code> , <code>READ:DEBUg:EVENT?</code> , <code>CONFIGure:DEBUg:IO</code> , <code>READ:DEBUg:IO?</code>
Log	<code>CONFIGure:LOG:UART</code> , <code>READ:LOG:UART?</code> , <code>READ:LOG:NEXT?</code> , <code>READ:LOG:ALL?</code> , <code>READ:LOG:STATUS?</code> , <code>CONFIGure:LOG:CLEAR</code>

IO Configuration and Revision History

PinProbe A1 Box Control SCPI Command Reference

Revision History

Date	New Commands	Revision
2026-09-24	CONFigure:IO:PROBe, IO TYPE commands, and output PULSe:WIDTh	Consolidated the day's IO update: added switchable RAW / FORWARD probing, persistent input/output LEVEL / PULSE configuration, common output pulse width, direction-scoped input latch/release paths, and the Tools Probe/Configure UI. Input latching is independent of Risk Mode; Risk Mode remains limited to pressure-sensor close confirmation, and existing applications remain LEVEL by default.
2026-09-12	CONFigure:MODE / READ:MODE? / CONFigure:MODE:ENABLe / READ:MODE:ENABLe?	Added the implemented and real-device-verified LOCAL / REMOTE / MIXED control modes, permission matrix, query and safety exceptions, Flash persistence for mode switching enable, recovery paths, and the distinction between -201, "Invalid while in local" and -224, "Illegal parameter value".
2026-08-11	CONFigure:DUT:AUTO / READ:DUT:AUTO? / READ:DUT:STATe?	Added DUT presence participation control for the USB automatic sequence, with Flash persistence. When USB:AUTO OFF, both USB automation and DUT detection are bypassed; when USB:AUTO ON, DUT detection can be enabled or disabled independently. The DUT state query always reads the actual sensor and is not bypassed.
2026-07-28	CONFigure:CYLInder2 / READ:CYLInder2:STATe?	Revised USB manual control and state return semantics: CLOSE/CLOSED indicates USB inserted/connected; OPEN/OPENED indicates USB unplugged/retracted; hardware output behavior is unchanged.
2026-07-27	CONFigure:USB:AUTO / READ:USB:AUTO?	Added automatic USB plug/unplug sequence commands; factory configured by device hardware, do not modify ; devices with USB cylinder are ON, devices without USB cylinder are OFF; customer software uses READ:USB:AUTO? to confirm whether enabled.